RADAR user guide

The air traffic around Helsinki-Vantaa as an air traffic control radar display

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In short. RADAR at `https://radar.oh2gax.org` shows the aircraft heard by the MODE-S Wind receiver in Tuusula as a controller would see them: on a radar scope, with three-line labels, speed vectors and history plots, in plain grey and black — colour only for alarms. Every aircraft is updated every second. The page shows:

- **the scope** — targets with speed vectors and trails, the **history line** of a selected aircraft, range rings, coastline and other map lines, airports and navaids, the airspace and the final approaches of the Finnish airports from the AIP;
- **the labels** — **Classic** or **TopSky** (as on a controller's display), and the six-line **extended label** with the aircraft's own airspeeds, heading, wind and temperature from its Mode S replies;
- **the alarms** — a red **GA** after a go-around at Helsinki-Vantaa, and the emergency squawks 7500 / 7600 / 7700 in red;
- **the windows** — the **TRAFFIC** list, **EFHK TRAFFIC** (landings and departures), **METAR & TAF** with the pilot **reports**, and What's new with the user guides;
- **replay** — any moment of the last **30 days**, played back at 1×–60×, with a link to share it;
- **the mobile app** — the same scope on a phone at `/app`.

Section 1 gets you started, Sections 2–3 explain the scope and the labels, Section 4 the left pane, Section 5 the windows, Section 6 the replay, Section 7 the phone. Section 8 tells where the data come from, Section 9 what to do when something looks wrong, and Section 10 is a quick reference.

Not for air traffic control or flight operations. RADAR is a hobby and research display. A passive receiver hears the aircraft, not the ground radars; an aircraft can be missing, late or wrong. Most figures of this guide show *simulated* traffic of the test system — the layout is the real one, the aircraft on your screen will differ.

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1 Getting started

Open `https://radar.oh2gax.org` in a browser on a computer (Chrome, Edge, Firefox or Safari; any screen from 1280 pixels wide works). No login is needed. The page has two parts (Fig. 1):

- the **left pane** with the settings, the UTC clock and the status line (Section 4);
- the **scope** — everything else: the radar picture and the windows you open on it (Sections 2, 3 and 5).

All your choices — the label style, the range, the centre, the theme, which windows are open and where they are — are remembered in your browser. RESET UI at the bottom of the pane brings everything back to the defaults. On a phone, open `radar.oh2gax.org/app` instead (Section 7).

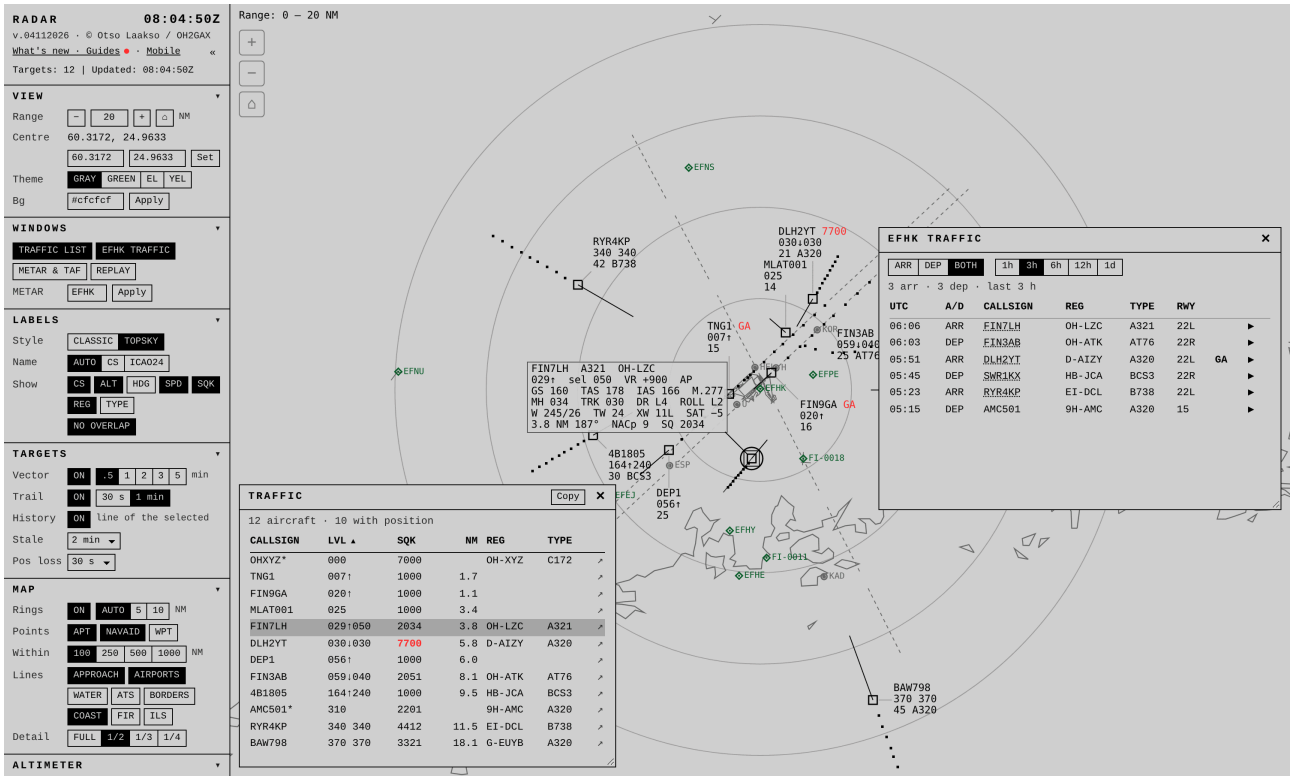


Figure 1: The page with TopSky labels, the TRAFFIC window (bottom left) and the EFHK TRAFFIC window (right). FIN7LH is selected: its extended label is open and its row in the TRAFFIC list is highlighted. DLH2YT squawks 7700 and TNG1 and FIN9GA have just gone around (red GA).

The status line at the top of the pane says how many targets there are and when the data were last updated (`Targets: 12 | Updated: 08:02:25Z`). If the receiver's feed stops, it turns red: `FEED LOST – no data for 45 s`; the targets are taken away after 30 seconds, so a dead feed is never shown as frozen aircraft. `Radar server not reachable` means that your browser cannot reach the server at all (your network, or the server is being updated).

All times are UTC (Finland: UTC + 2 h in winter, + 3 h in summer). The clock at the top of the pane shows the UTC time — during a replay, the replay time.

2 The scope

2.1 Moving and zooming

- **Drag** the map with the mouse to move it. ⌂ (top left of the scope, and in the pane's VIEW section) brings it back to the default centre, Helsinki-Vantaa.
- **Mouse wheel**: zoom in and out around the mouse pointer. **+** and **−** at the top left zoom around the centre. The range is 2–400 NM; the text `Range: 0 – 20 NM` at the top left shows the current one.
- The centre can also be typed in decimal degrees (VIEW → Centre → Set).

2.2 Targets, vectors and trails

Every aircraft with a position is a small **square**. The **speed vector** — a line from the square — shows where the aircraft will be after the vector time (0.5–5 minutes, TARGETS → Vector), so a longer line means a faster aircraft. The **trail** — small dots behind the square, one every 5 seconds — shows where it has been in the last 30 seconds or minute (TARGETS → Trail). An aircraft heard without a position (Mode S only) is not on the scope; it is listed in the TRAFFIC window with a ⋆ after its callsign.

2.3 The history line

Click an aircraft (with Classic or TopSky labels) to **select** it: a thin line shows where it has flown during this whole flight — the approach, a go-around, holding, the climb-out (Fig. 2). The line grows as the aircraft flies. The radar records every aircraft's positions itself and keeps them for 24 hours: straight flight about every 5 seconds, but every point while it turns, climbs or descends, so turns and go-arounds come out smooth. A part with no positions for over 2 minutes is drawn dashed. Click the aircraft again, click the empty scope or press **Esc** to remove the line. TARGETS → History switches the line off.

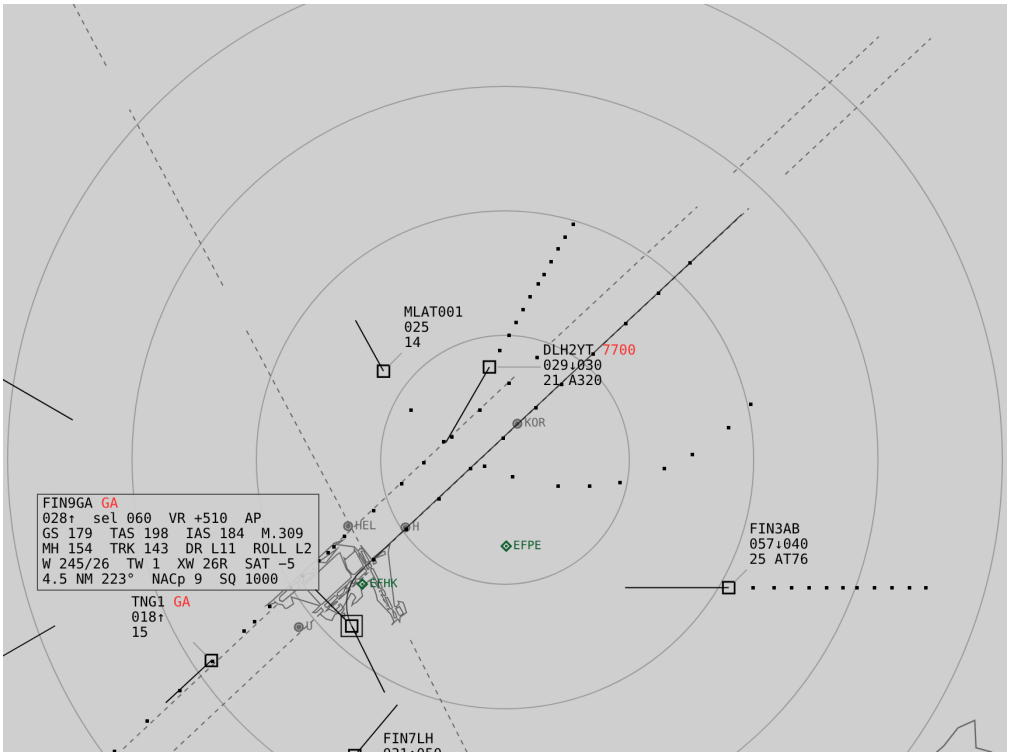


Figure 2: The history line of FIN9GA: the approach to runway 22L from the north-east, the go-around over the airport and the climb straight ahead. The extended label is open because it is selected. The dashed lines are the EFHK final approaches.

2.4 Range rings, map lines and navaids

- **Range rings** around the centre: AUTO draws four rings at a quarter, half, three quarters and the whole range; 5 or 10 draws a ring every 5 or 10 NM (MAP → Rings).
- **Airspace and routes** (MAP → Lines) come from the official **AIP Finland** and are updated automatically (Section 8): TMA/CTA — the TMAs, CTAs and CTRs, solid lines (on by default); FIR — the Finland FIR, dashed (on by default); SECTORS — the ACC sectors, dotted; P/R/D — the prohibited, restricted and danger areas, long dashes; TSA/TRA — the military training areas, dash-dot; ATS — the 28 RNAV routes (above FL 95 Finland is free route airspace, so there are few fixed routes) (Fig. 4).
- **Limits** (MAP → Limits) labels the airspaces as on a chart: the name and the airspace class, and under them a small box with the upper limit over the lower one (`FL285` over `2500`: from 2500 ft to FL 285). TMA/CTA labels the TMAs, CTAs and CTRs, SECTORS the ACC sectors (their letter) and P/R/D the prohibited, restricted and danger areas (identifier and limits); each set is labelled while its lines are on. A label sits where its area has the most room and only when the area is big enough on the screen — zoom in to see more of them.
- **Active areas** (MAP → Active) shows in **red** the areas that are active right now (Fig. 3): P/R/D — the restricted and danger areas switched on by NOTAM (firing ranges, UAV flights, exercises; a temporary area given only by a centre and a radius is a red circle), TSA/TRA — the military training areas booked in the day's Airspace Use Plan (AUP, and

its updates UUP). Each active area has a tag with its identifier, its limits and when it ends (`EFR92A / SFC–FL310 →1400Z`; a date when it ends on a later day). The radar reads both from AIS Finland every 30 minutes and follows the NOTAM schedules — an area booked `MON–FRI 0600–1500` is red only then. The number after the buttons says how many areas are active; its tooltip, when the sources were last checked. Inactive: SHOW draws the other areas of the set as usual when the set's Lines button is on, HIDE leaves only the active ones. In a replay the red areas are those that were active at the replay time.

- **Map lines**: APPROACH — the final approach of every runway with an instrument approach (ILS, LOC, RNP, VOR or NDB) at the Finnish airports, from the AIP: dashed extended centrelines from the runway threshold along the runway's bearing, 15 NM out with a gap at 5 and 10 NM, as on MODE-S Wind's ATC map — at Helsinki-Vantaa all six (04L / 04R / 22L / 22R / 15 / 33); AIRPORTS — runways and aprons; WATER — lakes and rivers; BORDERS; COAST — the coastline. Detail draws the lines with every point (FULL) or every second, third or fourth (faster on a slow computer).
- **Points** (MAP → Points): APT airports (green diamonds with the ICAO code), NAVAID VORs, DMEs and locators (grey circles; in Finland from the AIP), WPT the IFR waypoints of the AIP — the en-route points, and the aerodromes' own SID / STAR / approach points only when zoomed in to 25 NM or closer. They are loaded within the Within distance of the centre (100–1000 NM).

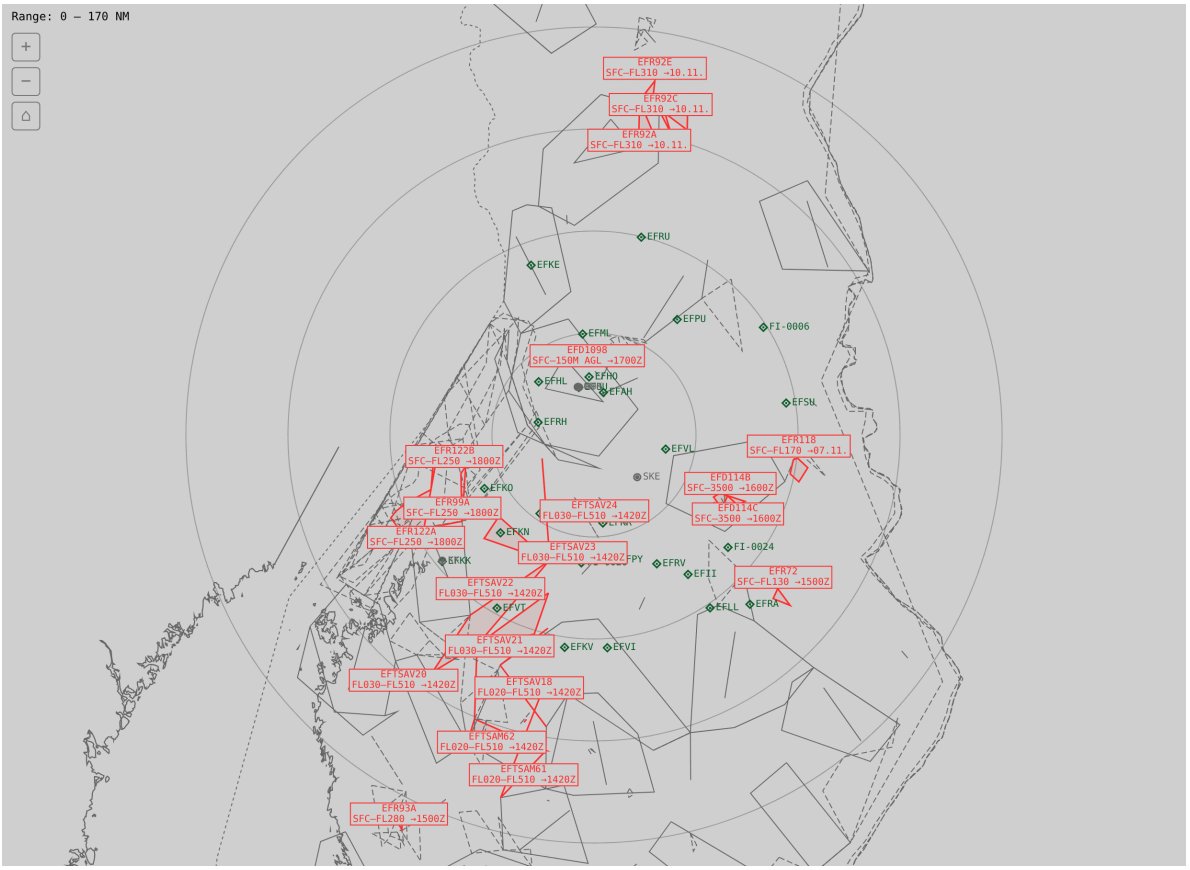


Figure 3: Active areas on 5 November 2026 at 13:30 UTC: the Rovajärvi firing range (EFR92A–E) in the north, the Lohtaja ranges (EFR99, EFR122) on the west coast, the training areas TSA V and TSA M booked in the AUP until 14:20, and the UAV area EFD1098 (a circle). P/R/D lines on, Inactive SHOW.

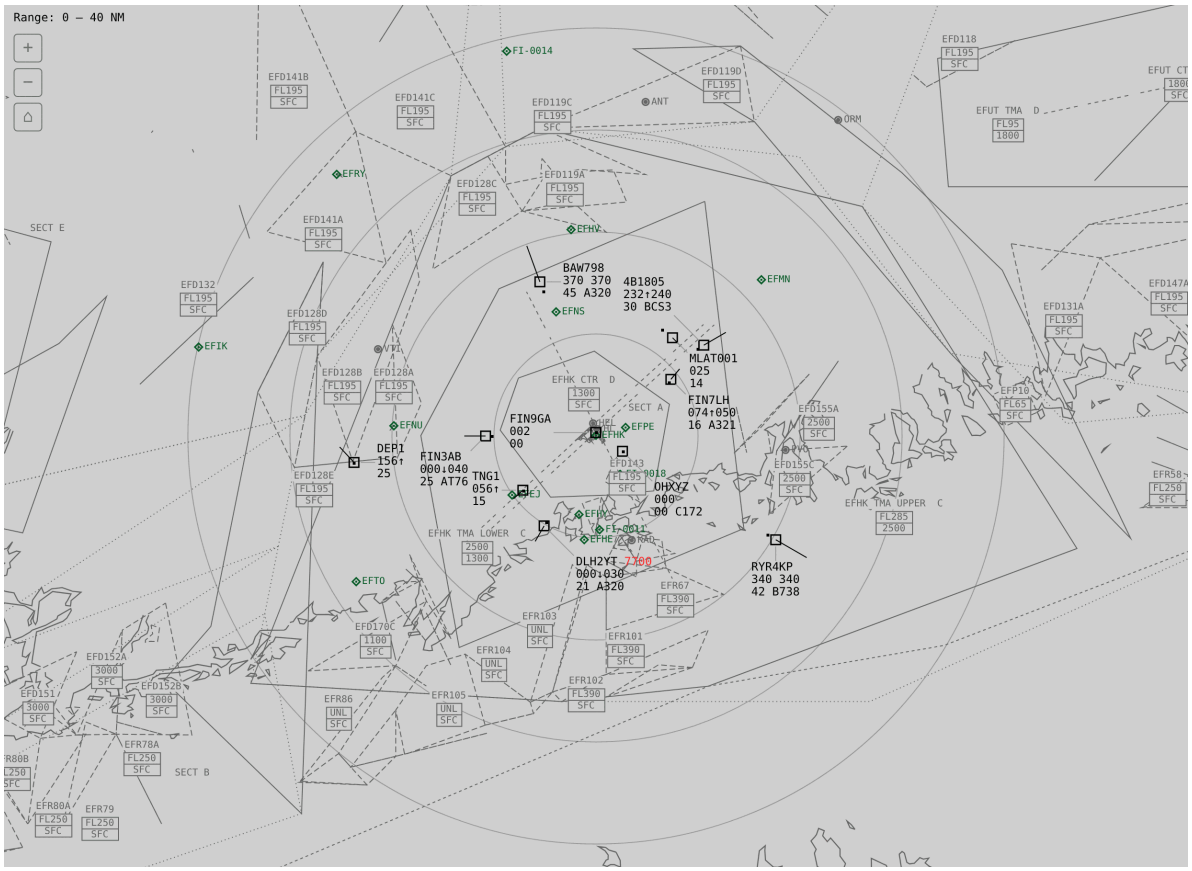


Figure 4: The airspace of AIP Finland around Helsinki-Vantaa at 40 NM with the Limits labels: the EFHK CTR and TMA (solid), the FIR (dashed), the ACC sectors (dotted) and the restricted and danger areas (long dashes), with the AIP's navaids.

2.5 Own position

OWN POSITION puts a small grey triangle at a place you type in decimal degrees (Apply, then SHOW) — your home, a spotting place. It is kept in your browser only. In the mobile app the phone's GPS gives it (Section 7).

3 The labels

3.1 Classic and TopSky

LABELS → Style chooses the label style (Fig. 5).

TopSky draws the labels as on a controller's radar display (the style takes its name from the TopSky ATC system): three lines and a thin **leader line** to the target.

1. **callsign** — or the ICAO24 address when the aircraft sends none — then the alarms in red (emergency squawk, GA);
2. **level**, **trend** and **selected level**, in hundreds of feet: `030↓030` = at 3000 ft, descending, cleared to 3000 ft; `164↑240` = climbing through FL164 to FL240; `340 340` = level at FL340. Below 5000 ft the level is the altitude above sea level (corrected with the QNH, Section 4.6), above it the flight level. ↑ / ↓ appear at 300 ft/min or more;
3. **ground speed** in tens of knots and the **aircraft type**: `21 A320` = 210 kt, Airbus A320.

The labels move by themselves so that they do not cover each other, the targets, the vectors, the trails, the navaid names, the approach lines or the windows, and they keep their place while it stays free. While you drag or zoom, they stay put and are placed again when the map stops.

Classic keeps the original labels of the radar: line 1 callsign and squawk; line 2 the level with `A` (altitude, below 5000 ft) or `F` (flight level), + climbing / − descending, the selected level (= when the aircraft is at it) and the ground speed in knots (`A025+ 140`, `F340=340 420`); line 3 the heading, registration and type, as chosen under LABELS → Show (CS ALT HDG SPD SQK REG TYPE). NO OVERLAP keeps Classic labels apart.

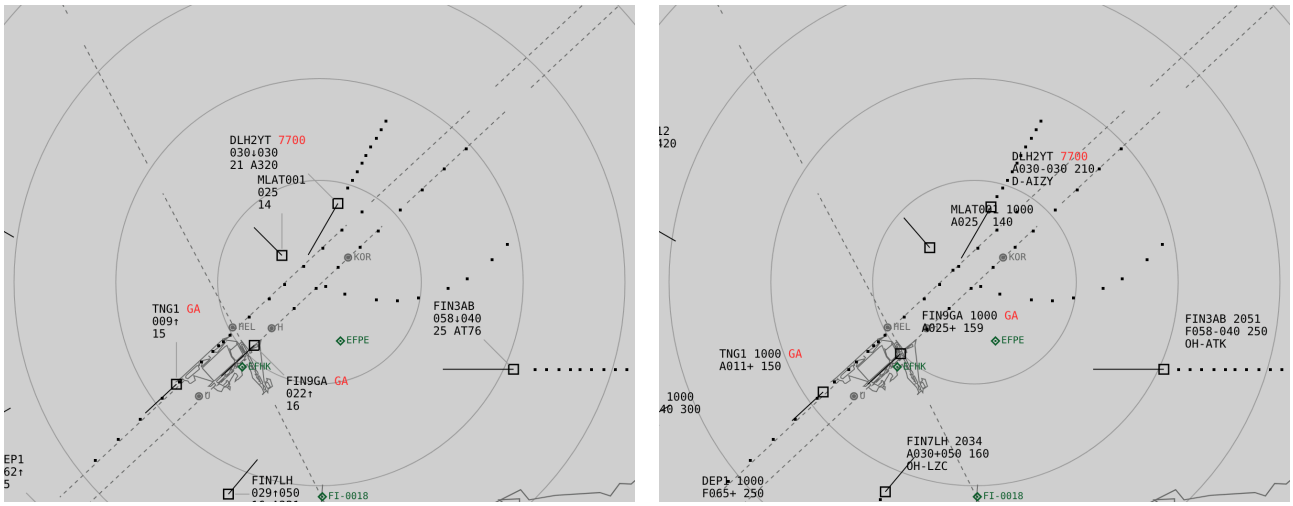


Figure 5: The same moment with TopSky labels (left) and Classic labels (right). DLH2YT squawks 7700; TNG1 and FIN9GA have gone around.

LABELS → Name chooses what line 1 shows: AUTO the callsign or, without one, the ICAO24 address; CS the callsign; ICAO24 always the address.

3.2 The extended label

With TopSky labels, move the mouse over an aircraft: its label grows to six lines (Fig. 6). Click the aircraft to keep it open; click it again or the empty scope to close it.

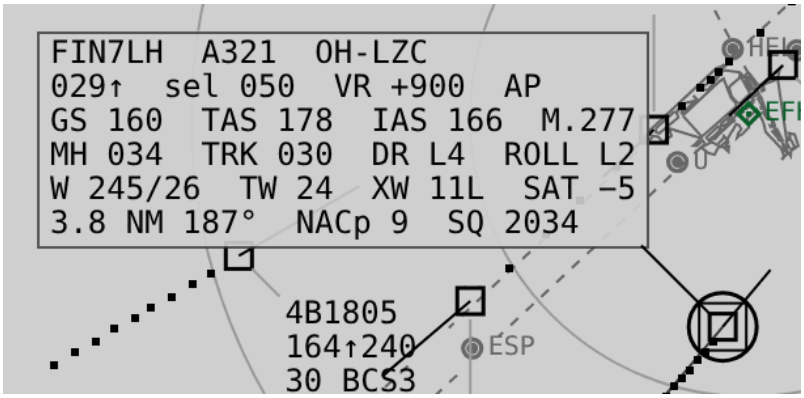


Figure 6: The extended label of FIN7LH.

Line	Contents
1	callsign, type, registration
2	level and trend, `sel` the selected level, `VR` the vertical rate (ft/min), `AP` / `APP` autopilot / approach mode engaged
3	`GS` ground speed, `TAS` true airspeed, `IAS` indicated airspeed (kt), `M` Mach number
4	`MH` magnetic heading, `TRK` magnetic track, `DR` drift (`L4` = track 4° left of the heading), `ROLL` bank angle
5	`W` wind direction / speed (kt), `HW` / `TW` its head- or tailwind and `XW` crosswind component on the track, `SAT` the outside air temperature (°C)
6	distance and bearing from the radar centre, `NACp` (the accuracy class of the ADS-B position, 9–11 = better than 30 m), `SQ` squawk

The air data come from the aircraft's own Mode S replies (Comm-B registers BDS 4,0, 5,0 and 6,0), received by MODE-S Wind's receiver and decoded by the radar; the wind and the temperature are averaged over the last minute. A value not heard for 20 seconds shows as `--`. Many aircraft — most turboprops and light aircraft, and any aircraft the ground radars do not ask — have no Comm-B values, and then lines 3–5 are mostly `--`.

3.3 The alarms

The radar mimics a real ATC display: grey and black, **colour only for alarms**.

- **GA — go-around.** When an aircraft goes around at Helsinki-Vantaa, a red `GA` appears after its callsign (TopSky and Classic), flashing for the first minute, then steady until it lands — or leaves the area, at most 30 minutes. A touch-and-go counts as a go-around too. The radar finds the go-around itself, within a few seconds, from the one-second positions (Section 8).
- **Emergency squawks** `7500` (unlawful interference), `7600` (radio failure) and `7700` (emergency) are red after the callsign, in the extended label and in the TRAFFIC window.

In the mono themes (Full EL) there is no red: the alarms are underlined or shown in the theme colour.

4 The left pane

The pane (Fig. 7) is drawn in the same style as the windows on the scope. An on/off setting is a small button that is **dark while it is on**; a row of joined buttons chooses one of a few (e.g. CLASSIC | TOPSKY). Each section opens and closes from its title (▼ / ▶); a closed one shows a short summary on the right (e.g. `Q0997 auto`). The pane remembers which sections are open. **«** at the top hides the whole pane for a full-width scope; **»** under the zoom buttons brings it back.

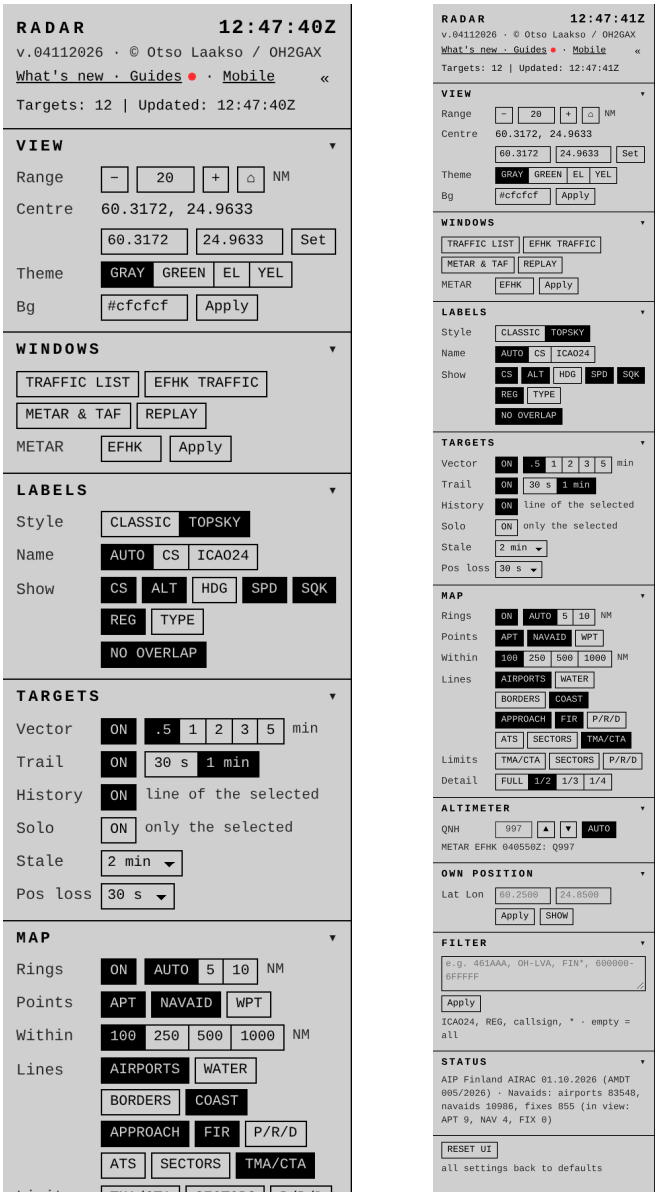


Figure 7: The left pane as it opens (left) and with every section open (right).

4.1 Header

RADAR and the UTC clock; the version (`v.DDMMYYYY`, the day of the latest change); What's new · Guides — a red dot after it marks a version you have not opened yet (Section 5.4); Mobile — the phone version; «; the status line (Section 1).

4.2 VIEW

Range − [20] + ⌂ NM — type a range or step it; ⌂ returns to the default centre. Centre — the current centre, and two boxes to type a new one (Set). Theme — GRAY (ATC grey, the default), GREEN (black with green, as an old electroluminescent display), EL (one dim green, mono) and YEL (one yellow, mono) — Fig. 8; Bg — the background colour of the Gray theme (e.g. `#cfcfcf`).

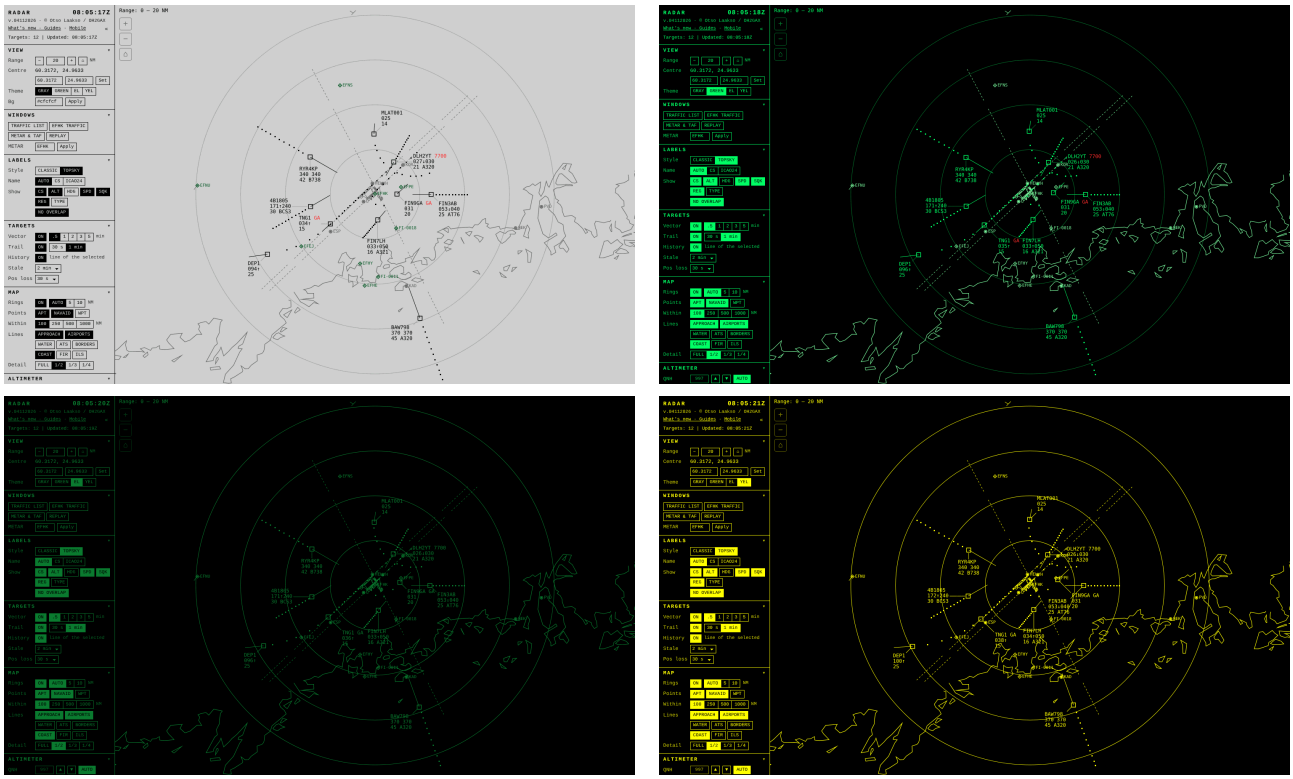


Figure 8: The four themes: Gray (ATC), Green (EL), Full EL (mono) and Full EL (yellow).

4.3 WINDOWS

Opens the windows on the scope (Section 5): TRAFFIC LIST, EFHK TRAFFIC, METAR & TAF (a red dot: a new pilot report near EFHK) and REPLAY (the replay bar). METAR — the airport of the METAR & TAF window (EFHK by default, any ICAO code such as EFTU).

4.4 LABELS

Style Classic / TopSky, Name AUTO / CS / ICAO24, Show what the Classic labels show (CS callsign, ALT level, HDG heading, SPD speed, SQK squawk, REG registration, TYPE type), NO OVERLAP — keep Classic labels apart (Section 3).

4.5 TARGETS

Vector on/off and its time (.5, 1, 2, 3, 5 min), Trail on/off and its length (30 s, 1 min), History the history line of the selected aircraft. Solo: while an aircraft is selected, **only it** is drawn — click it on the scope (or its row in the TRAFFIC window), and everything else disappears until you release it (click the empty scope or press Esc); the scope's corner says `SOLO: FIN7LH`. The TRAFFIC window still lists all aircraft, so a click on another row moves SOLO to that one. Stale: an aircraft whose position has not changed for this long (2 min by default) is removed. Pos loss: an aircraft whose position has not changed for this long (30 s) is moved off the scope to the no-position list (⋆ in the TRAFFIC window). Off keeps them.

4.6 MAP

Rings on/off and their spacing, Points APT / NAVAID / WPT, Within how far from the centre the points are loaded, Lines one button per set of map lines (APPROACH, AIRPORTS, WATER, ATS, BORDERS, COAST, FIR, P/R/D, SECTORS, TMA/CTA, TSA/TRA), Limits the labels of the airspaces (TMA/CTA, SECTORS, P/R/D), Active the active areas in red (P/R/D, TSA/TRA) and Inactive SHOW / HIDE, Detail (Section 2.4).

4.7 ALTIMETER

Below 5000 ft the levels are altitudes above sea level: the pressure altitude the aircraft sends is corrected with the **QNH** — 27 ft per hPa from 1013.25 hPa. With AUTO (on by default) the QNH comes from the latest METAR of EFHK, refreshed every 10 minutes; the METAR line under it shows which. Switch AUTO off to type a QNH or step it with ▲ ▼.

4.8 OWN POSITION, FILTER, STATUS

OWN POSITION — Section 2.5. FILTER — show only the aircraft you type, separated by commas or spaces, then Enter or Apply: an ICAO24 address (`461E1A`), a range of addresses (`460000-467FFF`: aircraft registered in Finland), a **registration** (`OH-LZC`; `OHLZC` works too) or a **callsign** (`FIN7LH`); ⋆ stands for any characters (`FIN⋆` all Finnair flights, `OH-⋆` all Finnish registrations). Empty = all aircraft. While it is on, the scope's corner says `FILTER ON`. STATUS — which AIRAC amendment of AIP Finland is in use (e.g. `AIRAC 01.10.2026 (AMDT 005/2026)`), and how many airports, navaids and waypoints are loaded and in view. Under it **DATA**: the downloads of the server — AIP: the AIRAC in use, when the AIP was last checked and the next check (once a day), and the next AIRAC amendment with the day it will be installed (AIS Finland lists it in advance); NOTAM and AUP: the sources of the active areas, last check, next check (every 30 minutes), when they last changed and how many activations or areas they have. A failed check is shown in red; the previous data are then still in use. The app has the same under ≡ → DATA. RESET UI at the bottom: every setting of this page back to the default — TopSky labels, the airspace lines with their limits, the active areas with the inactive ones hidden, 23 NM, and the TRAFFIC LIST, EFHK TRAFFIC and METAR & TAF windows open; a new browser starts the same way.

5 The windows

The windows open on the scope from WINDOWS in the pane. Each is moved by its title bar and resized from its lower-right corner; its place, size and settings are remembered. × closes it. The TopSky labels keep clear of the windows.

5.1 TRAFFIC

Every aircraft the radar hears, one row each (Fig. 9): CALLSIGN (⋆: no position), LVL as in the TopSky label, SQK (emergency squawks in red), NM the distance from the centre, REG and TYPE. The line above the table counts them.

- **Sort**: click CALLSIGN, LVL or NM; click again for the other way (▲ / ▼).
- **Click a row** to select that aircraft on the scope — its history line, and with TopSky its extended label; the row is highlighted. Click it again to release it.
- ↗ at the end of a row opens the aircraft on flightradar24 (by callsign) or ADS-B Exchange (by address) in a new tab.
- Copy in the title bar copies the list as text.

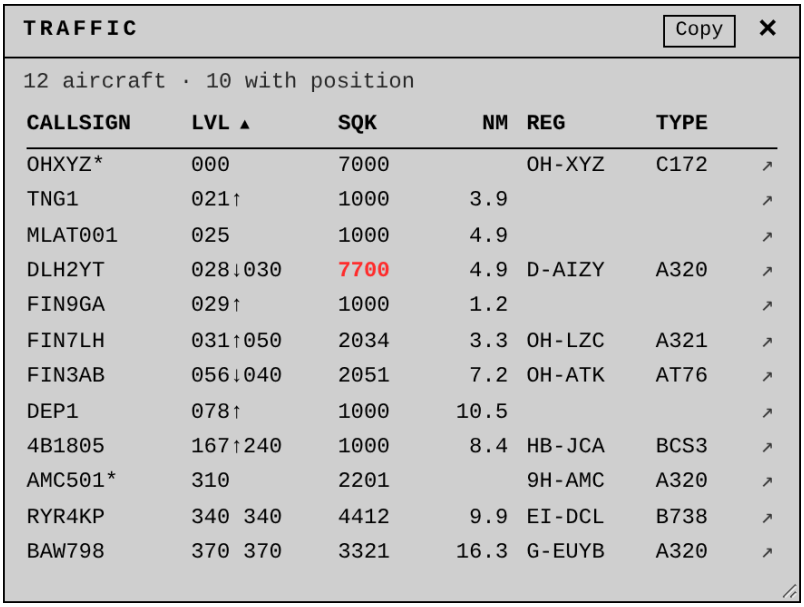


Figure 9: The TRAFFIC window, sorted by the level.

5.2 EFHK TRAFFIC

The landings and departures at Helsinki-Vantaa, newest first (Fig. 10): UTC time, callsign, registration — or, with ICAO24 instead of REG, the 24-bit address — type and runway. ARR, DEP or BOTH (then an A/D column tells which), and 1h … 1d; a line marks each change of the UTC day. A landing after a go-around is marked `GA` (`GA2` after two).

- The list comes from MODE-S Wind, which runs beside the radar: every landing on all runways, and the departures whose climb it measured. The radar also detects landings and departures itself and adds those MODE-S Wind does not have — mostly departures without Comm-B replies (the row's tooltip says *detected by the radar*).
- **Click a row** to draw that flight's track on the scope, also for an aircraft that has already gone, with a tag at its end (e.g. `DLH2YT 05:51 ARR 22L GA`); click the row again to remove it. An aircraft still on the scope (callsign underlined with dots) also blinks and is selected. Tracks are kept for 24 hours.
- ▶ replays that landing from 5 minutes before, or that departure from 2 minutes before, with the aircraft selected (Section 6).

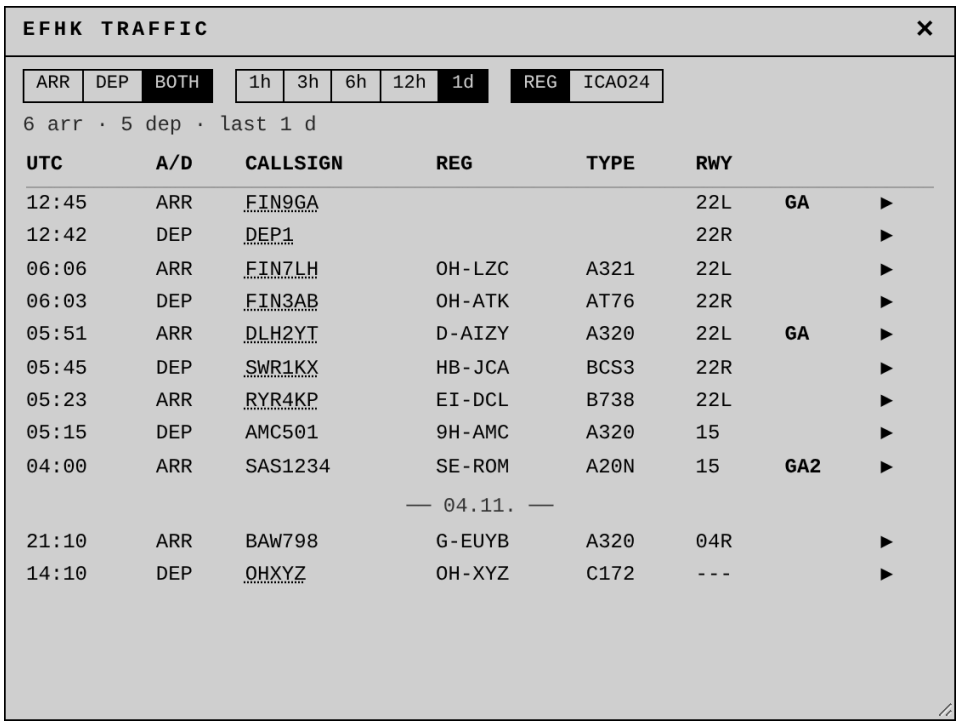


Figure 10: The EFHK TRAFFIC window: arrivals and departures of the last day, with the registrations (REG).

5.3 METAR & TAF and the pilot reports

The latest **METAR** and **TAF** of the airport chosen in WINDOWS → METAR, with their times and how old they are; a METAR older than 90 minutes is red (Fig. 11). They come from the same NOAA source as in MODE-S Wind and are refreshed every 10 minutes.

The second tab, REPORTS, lists the **pilot reports** that MODE-S Wind collects from the Finnish aviation weather service every 30 minutes: `ARS` and `AIREP` (special air-reports: moderate or severe turbulence or icing, mountain waves, thunderstorms …), the national `WXREP` (light turbulence or icing, windshear, inversions, CB tops), `SIGMET` and `AIRMET`, and the EFHK **METARs with a windshear group** (`WS R22L`).

- 12 h / 24 h; ALL FINLAND or NEAR EFHK (within 30 NM of the airport or naming it, at or below 10 000 ft, tagged `EFHK`); which kinds to show.
- Severe (`SEV`) and heavy (`HVY`) phenomena are red.
- ARS on scope marks the reports with a position on the scope: a small diamond with the phenomenon and level.
- A red dot on the tab, on METAR & TAF in the pane and on the app's menu tells that a report near EFHK has arrived since you last looked.

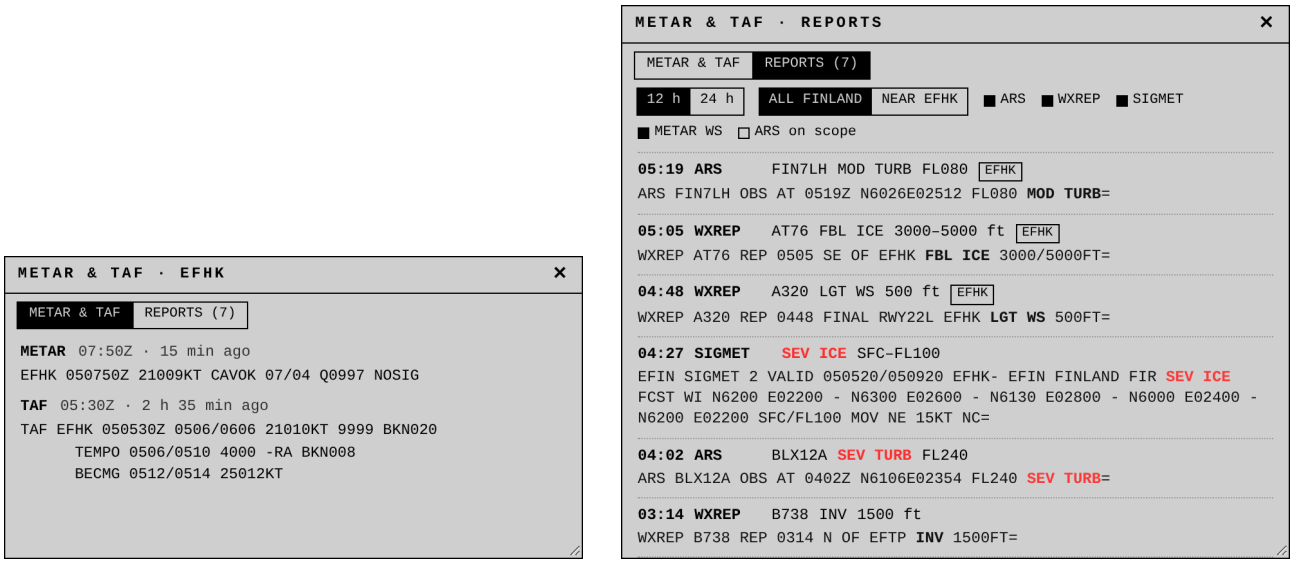


Figure 11: The METAR & TAF window: the METAR and TAF of EFHK (left) and the pilot reports (right).

5.4 What's new and the user guides

What's new · Guides in the pane's header opens a window with three tabs: WHAT'S NEW — the changes of the ten newest versions, in plain words —, USER GUIDES — this guide and any later ones, with Open PDF and Download (Fig. 12) — and ABOUT — what RADAR is and where its data come from. The window moves and resizes like the others and stays open while you use the scope; × or Esc closes it.

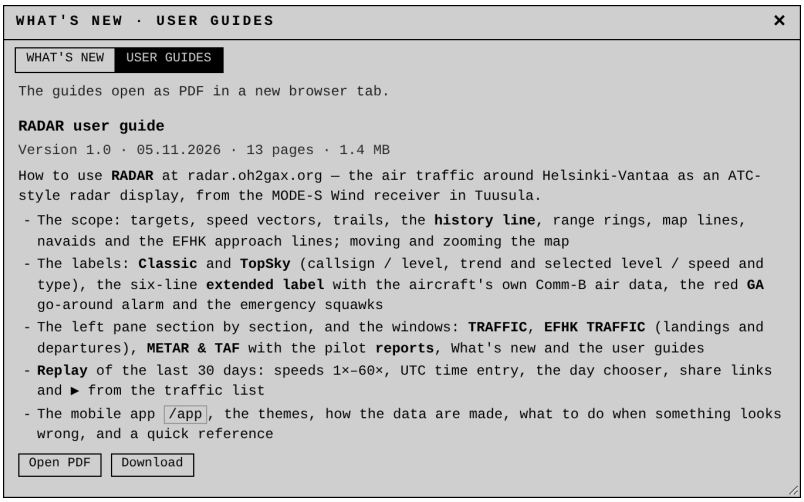


Figure 12: The USER GUIDES tab of the What's new window.

6 Replay

The radar records everything it shows, every second, and keeps it for **30 days**: every aircraft within 150 NM of Helsinki-Vantaa with its labels and the extended label's values, the go-arounds, landings and departures, and the METARs. Any moment of these days can be played back on the scope exactly as it looked live (Fig. 13).

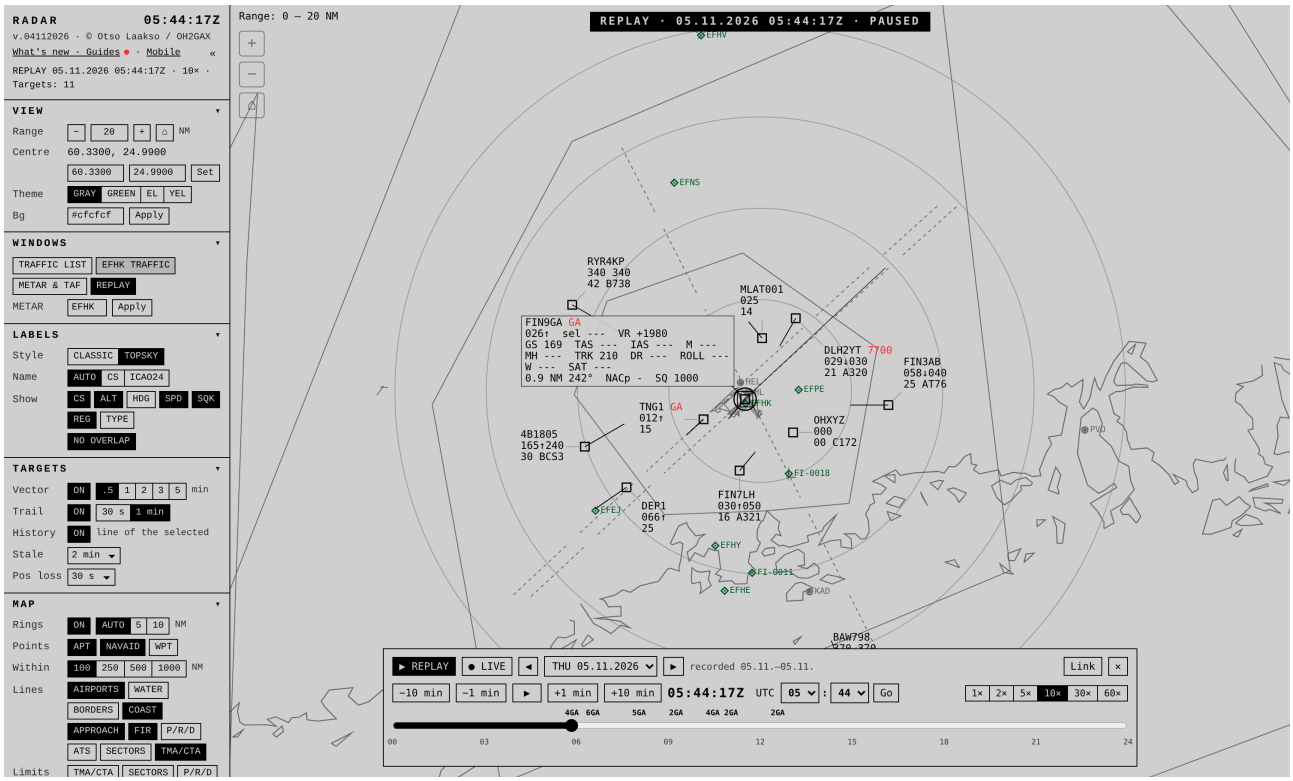


Figure 13: A replay, paused: the banner at the top, the replay bar at the bottom. FIN9GA is selected; its history line ends at the replay time.

Open the replay bar from WINDOWS → REPLAY. While a replay runs, the inverted banner at the top of the scope says `REPLAY · 05.11.2026 05:44:17Z · 10×` (or `PAUSED`), so a replay is never mistaken for live traffic. The clock in the pane shows the replay time.

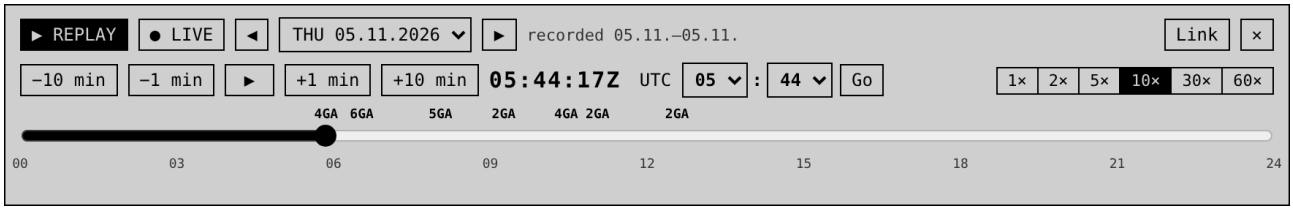


Figure 14: The replay bar.

The bar (Fig. 14):

- row 1: ▶ REPLAY starts the replay, ● LIVE returns to the live traffic at once; the **day** (◀ previous, the list of the last 30 days — days without a recording are greyed — ▶ next); what has been recorded; Link; × closes the bar;
- row 2: −10 min, −1 min, play / pause (also the **space bar**), +1 min, +10 min; the replay time; the **UTC time** — choose the hour and the minute from the two lists HH : MM and press Go to jump there on the chosen day (paused); the lists show the replay time until you change them; the speeds 1× 2× 5× 10× 30× 60×;
- row 3: a slider over the whole UTC day; the day's go-arounds are marked on it with `GA` — point at one to see which.

In a replay the labels, the extended label, the GA alarm (from the recorded go-arounds), the trails and vectors work as live; the history line shows the flight up to the replay time; with AUTO the QNH is that of the METAR of that time. Link copies a link to this moment with the map's centre, range and the selected aircraft — anyone who opens it sees the same replay, e.g. `https://radar.oh2gax.org/?replay=2026-11-05T05:44:17Z&speed=10&c=60.33,24.99&r=20&sel=461AAA`.

The ▶ in the EFHK TRAFFIC window starts a replay a few minutes before that landing or departure. The replay plays at once at the speed chosen last, so at 30× or 60× the take-off may already be over when you find the aircraft — step back a minute (−1 min) or choose a slower speed. An aircraft on the ground is not always heard by the receiver; it may appear only a few hundred feet up.

7 RADAR on a phone

Open `radar.oh2gax.org/app` in the phone's browser (or Mobile in the pane). The scope fills the screen; every choice is in the ≡ menu, in the ATC grey style (Fig. 15). *Add to Home Screen* (Android: the browser's ⋮ menu; iPhone: the Share button) gives it its own icon, and it opens like an app.

- **One finger** moves the map, **two fingers** pinch to zoom. **Tap** an aircraft (or its label) to select it — TopSky's extended label and the history line — and tap the empty scope to release it. + / − / ⌂ at the bottom right.
- ⊙ shows your own position from the phone's GPS and centres the map on it; tap again to switch it off. The position is not stored or sent anywhere.
- The menu: the views (Radar, Replay, EFHK traffic with REG / ICAO24, METAR & TAF with the reports, Full version), the label style and what the labels show, vector, trail, history line and Solo, the map lines with the active areas, the FILTER (as on the desktop), the theme, Auto QNH, *Keep the screen on*, and What's new · Guides — the notes and the user guides.
- The app keeps its own settings: changing something on the phone does not change the desktop page in the same browser, and Reset in the app clears only the app's.

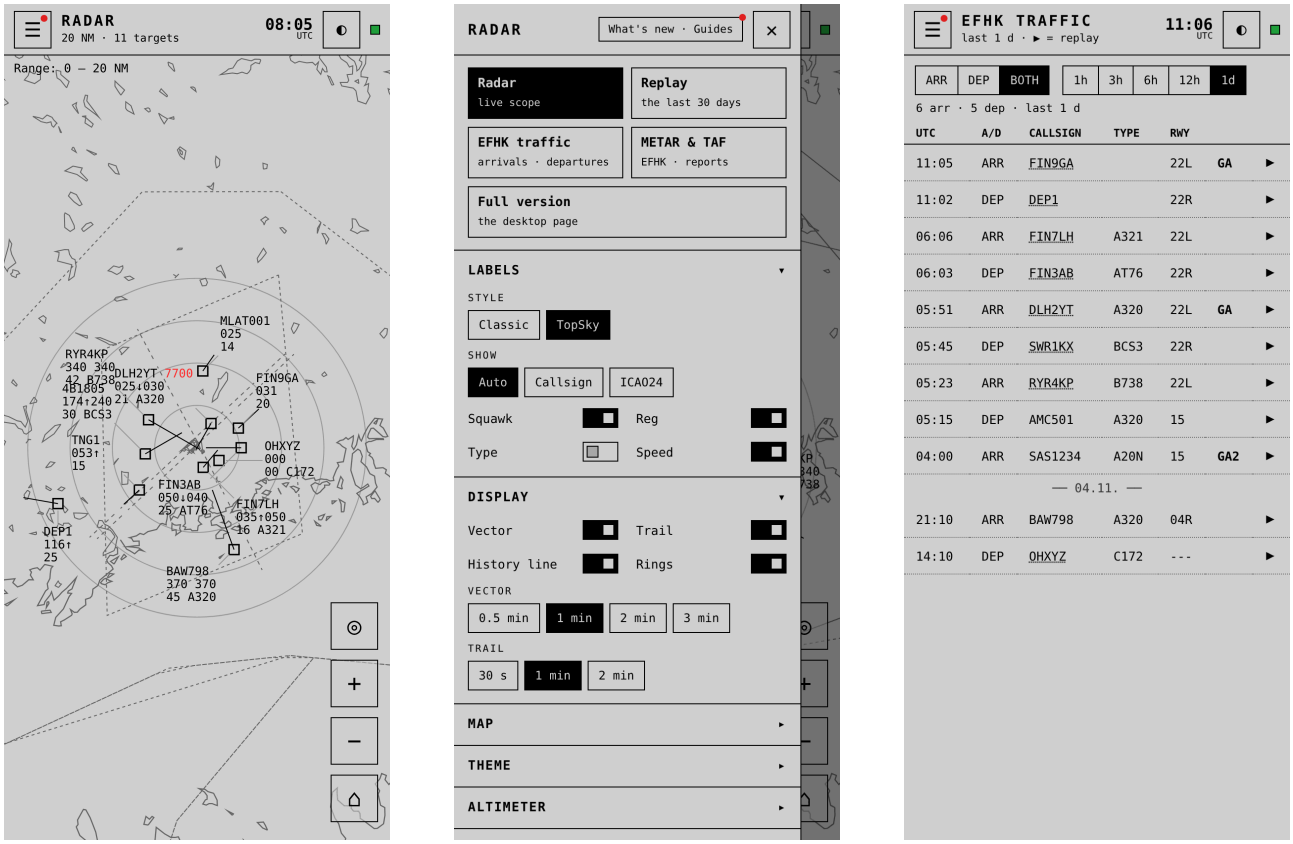


Figure 15: The mobile app: the scope, the ≡ menu and the EFHK traffic page (▶: replay that flight).

7.1 Replay on the phone

Replay in the ≡ menu opens the replay controls as a sheet at the bottom of the scope (Fig. 16); tap the tile again, or ×, to close it and go back to live. While a replay runs the bar at the top is **black** and says `REPLAY` with the replay time and speed (`05.11.  05:44:19Z · 10×` or `PAUSED`); the clock shows the replay time too.

- row 1: the replay time; Share — the phone's share menu (or a copied link) for this moment, the view and the selected aircraft; • LIVE back to the live traffic; × closes the sheet;
- row 2: the **day** (◀, the list of the last 30 days, ▶) and the **UTC time**: tap the hour and the minute (HH : MM open as the phone's scroll wheels) and tap Go;
- the slider of the whole UTC day, the go-arounds marked `GA` (several close together: `3GA`);
- −10′ −1′ ▶ +1′ +10′ as big buttons, and the speeds 1×–60×.

On the EFHK traffic page every row has a ▶: it replays that landing (from 5 minutes before) or departure (2 minutes before) with the aircraft selected. During a replay a tap on any row does the same. A replay link — `radar.oh2gax.`

`org/app?replay=...` from the app, `/?replay=...` from the desktop page — opens that moment on the page it was made on. The zoom buttons move up above the sheet; in landscape the sheet is on the left.

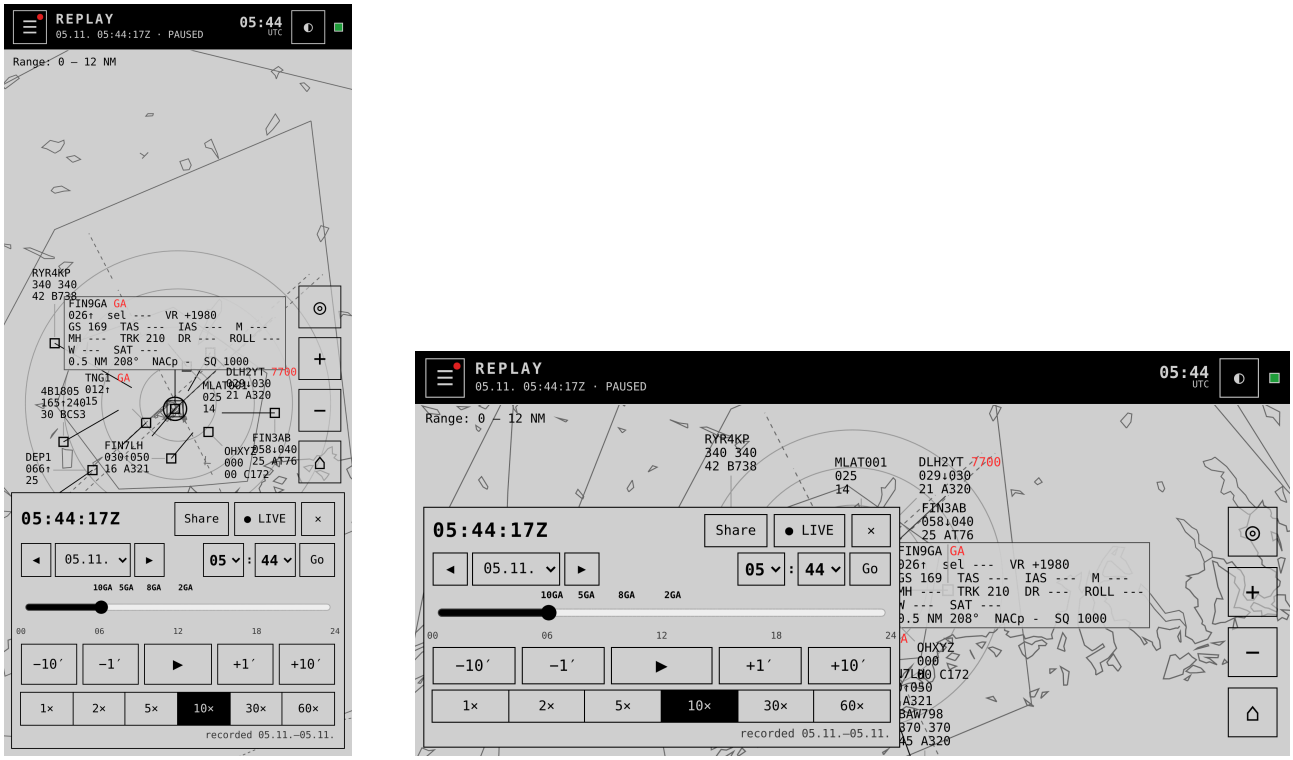


Figure 16: Replay in the app, paused just after FIN9GA's go-around: the replay sheet in portrait (left) and in landscape (right).

8 Where the data come from

- **The receiver.** MODE-S Wind's 1090 MHz receiver in Tuusula (a Jetvision Radarcape), just north of Helsinki-Vantaa, hears the replies and broadcasts of the aircraft's transponders — not the ground radars. It hears aircraft at cruise levels far away, but low-flying aircraft only much closer.
- **Positions every second.** The ADS-B positions are decoded from the receiver's raw message stream every second; aircraft without ADS-B positions (multilateration, MLAT) take the receiver's aircraft list, refreshed every 2 seconds.
- **Labels.** Callsign, squawk, level and vertical rate from the transponder; the selected level from the ADS-B or Comm-B (BDS 4,0) replies; the registration and type from the receiver's aircraft database, completed from a public database (tar1090-db).
- **Extended label.** The Comm-B registers BDS 5,0 (track, ground speed, true airspeed, roll) and 6,0 (magnetic heading, indicated airspeed, Mach) of the aircraft's replies to the ground radars; the wind from the difference of the air and the ground vectors, the temperature from the Mach number and the true airspeed — as in MODE-S Wind.
- **Go-arounds, landings and departures** are found by the radar every second, with the same rules as in MODE-S Wind: an aircraft established on an approach to an EFHK runway (in the ILS corridor, descending) that starts to climb — at least 600 ft/min below 2200 ft — or climbs away after crossing the threshold is a go-around; one that crosses the threshold and does not climb is a landing; one climbing out from the runway area below 1500 ft at 90 kt or more is a departure. The levels are corrected with the QNH of the latest EFHK METAR.
- **EFHK TRAFFIC**, the **reports** and the windshear METARs come from MODE-S Wind's database on the same server; the METAR and TAF from NOAA.
- **Navaids, waypoints, routes and airspace in Finland** come from the eAIP of AIS Finland (Fintraffic ANS, www.ais.fi). The radar checks once a day which AIRAC amendment is in force (Finland publishes about six a year) and installs a new one the day it takes effect — after checking that it is complete; if anything is wrong, the previous data stay. Outside Finland the navaids are from the OurAirports database. For information only, not for navigation.
- **Active areas** come from AIS Finland too (www.ais.fi): the NAV WRNG pre-flight information bulletin of the Helsinki FIR (the NOTAMs that activate restricted and danger areas) and the Airspace Use Plan AUP / UUP of the Airspace Management Cell (the training areas, valid 0600–0600 UTC). The radar reads them every 30 minutes and keeps every change for 31 days, for the replay. An area booked only in a NOTAM text without coordinates, or one not in the AIP, cannot be drawn.
- **The recordings**: the history lines 24 hours, the replay 30 days (150 NM), the active areas 31 days.

9 When something looks wrong

What you see	What to do
`FEED LOST` (red)	The receiver's feed has stopped (power, network or an update at the receiver). Wait; the scope fills again by itself.
`Radar server not reachable`	Your network, or the server is being updated (a minute). Reload the page if it lasts.
No aircraft at all, no warning	The FILTER may be on (the scope's corner says `FILTER ON`) or the range very small; RESET UI if unsure.
Only one aircraft on the scope	SOLO is on (TARGETS → Solo) and that aircraft is selected: click the empty scope or press Esc.
An aircraft jumps or is missing	A passive receiver misses replies, especially low down and far away; an MLAT position is coarser than ADS-B.
`--` in the extended label	The aircraft sends no Comm-B values, or none for 20 seconds (turboprops, light aircraft).
Levels below 5000 ft look 100–300 ft off	Check the QNH (ALTIMETER): AUTO uses the latest EFHK METAR.
A label shows the ICAO24 code	The aircraft sends no callsign (AUTO uses the address).
The page looks old after an update	Reload it (Ctrl+F5 / Cmd+Shift+R).
A window is gone	Open it again from WINDOWS; it comes back inside the scope. RESET UI puts everything back to the defaults.
The pane is gone	» under the zoom buttons brings it back.

10 Quick reference

Item	Meaning
Drag / wheel / + − ⌂	move / zoom around the pointer / zoom / back to EFHK (2–400 NM)
Click an aircraft	select: history line (and the TopSky extended label); again, the empty scope or Esc: release
Square, line, dots	target, speed vector (vector time), trail (a dot every 5 s)
`030↓030` / `21 A320`	TopSky: level, trend (↑ ↓ at 300 ft/min), selected level / speed in tens of kt and type
`A025+ 140`, `F340=340`	Classic: A altitude (below 5000 ft, QNH) or F flight level, + / − trend, = at the selected level, speed in kt
Red `GA`	go-around at EFHK: flashing 60 s, then steady until the landing (max. 30 min)
Red `7500 7600 7700`	emergency squawks
Dashed lines	the final approaches of the Finnish airports (15 NM, gaps at 5 and 10 NM); the FIR
Solid / dotted / long-dash	TMA, CTA, CTR / ACC sectors / P, R, D areas (AIP Finland)
AIRAC in STATUS	the AIP Finland amendment in use, updated automatically
Name and box `FL285 / 2500`	an airspace (MAP → Limits): class, upper limit over lower limit
⋆ after a callsign	no position (TRAFFIC window)
Dark button	on
« / »	hide / show the left pane
Red dot	What's new: an unopened version; METAR & TAF: a new pilot report near EFHK
TRAFFIC	sort by CALLSIGN / LVL / NM, row = select, ↗ flightradar24 / ADS-B Exchange
EFHK TRAFFIC	ARR / DEP / BOTH, 1 h–1 d, REG / ICAO24, row = the flight's track, ▶ = replay it
FILTER, SOLO	ICAO24, range, REG or callsign, ⋆ = any; SOLO: only the selected aircraft
Active	MAP → Active P/R/D (NOTAM), TSA/TRA (AUP / UUP): red, with limits and end; Inactive SHOW / HIDE
REPORTS	ARS, AIREP, WXREP, SIGMET, AIRMET, METAR WS; SEV / HVY red
Replay	30 days, 150 NM; space = play / pause, UTC HH : MM + Go, 1×–60×, Link; app: ≡ → Replay
`/app`	the phone version, own settings, ⊙ own GPS position
Settings	remembered in your browser; RESET UI clears them
Times	UTC everywhere

MODE-S Wind (`https://mode-s-wind.oh2gax.org`), which runs the receiver and the weather from the aircraft, has its own user guides in its What's new window.